

UHP Laser Sources for CPO

Source ID: CMP-016

Canonical original: <https://www.lumentum.com/en/products/data-center/cw-lasers/uhp-lasers-cpo>

Local status: readable text snapshot generated from retained publisher HTML on 2026-08-07.



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UHP Laser Sources for CPO

Ultra-high-power lasers enabling scalable pluggable external light sources for co-packaged optics.

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Lumentum's ultra high power (UHP) lasers deliver the optical power, spectral purity, and stability required for next-generation co-packaged optics (CPO) platforms in AI and cloud data centers. Operating at 1311 nm, these lasers achieve output powers up to 350 mW at 50°C and 235 mW at 70°C, with power-conversion efficiency above 20%. Their narrow linewidth (<500 kHz) and low relative intensity noise ($RIN < -147$ dB/Hz) ensure exceptional signal integrity for dense optical interconnects. Designed and manufactured on Lumentum's indium phosphide (InP) platform, the UHP laser combines proven reliability, compact packaging, and high manufacturing scalability for deployment in thermally demanding environments.

These UHP lasers are integrated into Lumentum's external laser source form factor pluggable (ELSFP) modules to provide a centralized, serviceable light source for silicon photonics (SiPh) optical engines in CPO systems. An ELSFP-based system architecture moves the laser function away from the GPU or ASIC, reducing heat inside the package, simplifying service and upgrades, and enabling multiple optical engines to share a single laser module. This centralized design lowers total cost of ownership, improves thermal management, and enhances system reliability—delivering a scalable, energy-efficient light source that enables the next generation of high-bandwidth CPO-based networking and compute architectures.

Product Resources



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